

Nicolás Alejandro Comay

PhD candidate

Psychological Research Institute (UNC-CONICET, Argentina)

✉ nicocomay@gmail.com 🌐 nicolascomay.github.io | Updated: Aug. 26, 2026

Education

PhD in Neuroscience, National University of Córdoba (UNC-Argentina).

2021-present

Supervisor: Dr. Pablo Barttfeld. Co-supervisor: Dr. Guillermo Solovey.

Degree in Psychology, National University of Córdoba (UNC-Argentina).

2016-2020

Recipient of the 'University Award' for academic excellence (highest GPA among on-time graduates).

Publications

Comay, N.A.; Solovey G.; Barttfeld, P. Confidence phenotypes: a unified computational account of value and decision certainty in reinforcement learning. *eLife*, 15:RP111820. 2026

Comay, N.A.; Solovey, G.; Barttfeld, P. Iconic vs. working memory metacognition to evaluate the richness of perception: a registered report. *Royal Society Open Science*, 12(11), 231805. 2025

Comay, N.A.; Solovey, G.; Barttfeld, P. Decisions are based on less information than metacognitive judgments in multialternative contexts. *Journal of Experimental Psychology: Learning, Memory and Cognition*, 52(7), 1080–1093. 2025

Barttfeld, P.; **Comay, N.A.**; Embon, I.; Solovey, G. Pre- and post-decision signals of certainty in changing minds. *Trends in Neurosciences*, 48(12), 935–937. 2025

Zapata, J. M.⁺, **Comay, N.A.**⁺, Taricco, G., Barttfeld, P., Solovey, G., Saal, A., & Ahumada, J. V. Metacognitive sensitivity on the Iowa Gambling Task reveals awareness as a necessary condition for advantageous performance. *Experimental Psychology*, 71(6), 343–351. ⁺joint first authors. 2025

Comay, N.A., Della Bella, G., Lamberti, P., Sigman, M., Solovey, G., & Barttfeld, P. The presence of irrelevant alternatives paradoxically increases confidence in perceptual decisions. *Cognition*, 234, 105377. 2023

Scholarships, awards and research funding

FELLOWSHIPS

PhD fellowship. National Scientific and Technical Research Council of Argentina (CONICET). 2021

Stimuli to Scientific Vocations scholarship. Undergraduate scholarship from the National Interuniversity Council of Argentina. 2020

AWARDS

Dr. Pedro Pavesi award for best presentation at the conference The Future of Decision-Making Sciences (Center of Rationality, Action and Decision – Economics School – University of Buenos Aires). 2022

University Award from the National University of Córdoba for academic excellence (highest GPA among on-time graduates). 2021

RESEARCH FUNDING

Pre-stimulus neural activity and conscious perception: a theory-informed investigation using report and no-report fMRI paradigms. *Consciousness Commons 2026 prize*. USD 8,960. 2026

Joint project by four PhD candidates; co-recipients of the award.

Incubator project prize, First Annual Summer School on Consciousness and Metacognition (University College London and Université de recherche Paris Sciences et Lettres). GBP 250. 2021

Joint project by four PhD candidates; co-recipients of the award.

TRAVEL GRANTS FOR ACADEMIC EVENTS

Travel grant to attend the 2025 Multi-disciplinary Conference on Reinforcement Learning and Decision Making (Trinity College Dublin, Ireland) **2025**

Travel grant to attend the Neuroscience of Consciousness Winter School (Canadian Institute for Advanced Research, CIFAR). **2024**

Travel grant to attend the RIKEN Center for Brain Science Summer School (Tokyo, Japan) **2023**

EMBO travel grant to attend the Barcelona Summer School for Advanced Modeling of Behavior (Centre de Recerca Matemàtica – Barcelona, Spain). **2022**

Research experience

AS A GRADUATE STUDENT

Three-month research internship at the Laboratory for Imagination and Executive Functions led by Dr. Kentaro Miyamoto (*RIKEN Center for Brain Science* – Hirosawa, Wako, Saitama 351-0106, Japan). **2024**

Two-month research internship at the Laboratory for Imagination and Executive Functions led by Dr. Kentaro Miyamoto as part of the *RIKEN Center for Brain Science Summer School Program* (Hirosawa, Wako, Saitama 351-0106, Japan). **2022**

Ph.D. candidate in Neuroscience at the Psychological Research Institute (National Scientific and Technical Research Council – National University of Córdoba, Argentina). **2021-present**

AS AN UNDERGRADUATE STUDENT

Undergraduate research assistant, supported by the *Stimuli to Scientific Vocations Scholarship* from the National Interuniversity Council of Argentina. Project: *Effects of pre- and/or postnatal ethanol exposure on the modulation of the opioid system in neonatal rats*. **2020-2021**

Thesis in Psychology (Psychological Research Institute, UNC-CONICET). Title: *Confidence in perceptual decisions with multiple alternatives*. Qualification: 10/10. Supervisor: Dr. Pablo Barttfeld. Co-supervisor: Dr. Guillermo Solovey. **2020**

Undergraduate research assistant. Project: *Neuroprotection facilitated by Omega 3 in two cognitive impairment models*. Psychological Research Institute (UNC-CONICET, Argentina). **2019**

Presentations

ORAL PRESENTATIONS & TALKS

Confidence phenotypes: a unified computational account of value and decision certainty in reinforcement learning (**Comay, N.A.**; Solovey, G.; Barttfeld, P.). 48th Annual Meeting of the Cognitive Science Society. Rio de Janeiro, Brasil. **2026**

Iconic vs. working memory metacognition to evaluate the richness of perception: a registered report (**Comay, N.A.**; Solovey, G.; Barttfeld, P.). 30th annual meeting of the Association for the Scientific Study of Consciousness. Santiago, Chile. **2026**

Bayesian approach to perceptual decision making and confidence (**Comay, N.A.**; Solovey, G.; Barttfeld, P.). Bayesian Plurinational Congress. Santiago del Estero, Argentina. **2023**

Computational modeling of behavior (**Comay, N.A.**; Solovey, G.; Barttfeld, P.). V International Psychology Congress and VIII National Psychology Congress “Science and Profession” (UNC, Argentina). **2023**

Deviations from Bayesian optimality in the evaluation of perceptual decisions (**Comay, N.A.**; Della Bella, G.; Lamberti, P.; Sigman, M.; Solovey, G.; Barttfeld, P.). The future of decision-making sciences conference (Center of Rationality, Action and Decision - Economics School – University of Buenos Aires). **2022**

Implausible alternatives paradoxically increase confidence in a perceptual decision (**Comay, N.A.**; Della Bella, G.; Lamberti, P.; Sigman, M.; Solovey, G.; Barttfeld, P). XXXVI Annual Meeting of the Argentinian Society for Neuroscience Research. **2021**

WORKSHOPS & SYMPOSIA

Thinking how we think: actual controversies in metacognition and experimental designs. (Embon, I.⁺; Lillo, M.⁺; **Comay, N.A.**⁺; Aguirre, J.⁺). II International Meeting Frontiers in Neurosciences (Universidad Nacional Autónoma de México, UNAM). ⁺equal contribution. **2023**

Reflecting upon our own thoughts: recent studies on metacognition in vigilance, personality and decision making (**Comay, N.A.**⁺; Aguirre, J.; Embon, I.; Gerbaudo, M.A.; Lillo, M.). XIX National Meeting and VIII International Meeting of the Argentinean Association for Behavioral Sciences.⁺symposium coordinator **2023**

POSTER PRESENTATIONS

Bayesian-like computations explain human confidence in values and decisions during learning (**Comay, N.A.**, Solovey, G.; Barttfeld, P.). 2025 Multi-disciplinary Conference on Reinforcement Learning and Decision Making. Trinity College Dublin, Ireland. **2025**

Computational modeling of uncertainty in reinforcement learning (**Comay, N.A.**, Solovey, G.; Barttfeld, P.). I Congreso Argentino de Ciencias Cognitivas. Universidad Torcuato Di Tella, Buenos Aires. **2024**

Implausible alternatives paradoxically increase confidence in a perceptual decision. (Comay, N.A.; Della Bella, G.; Lamberti, P.; Sigman, M.; Solovey, G.; Barttfeld, P.) Computational Cognition Workshop (Osnabrück, Germany). **2021**

Effect of subliminal feedback on metacognition in a visual perception task. (Zapata, J.M.⁺; **Comay, N.A.**⁺; Della Bella, G.⁺). XXXV Annual Meeting of the Argentinian Society for Neuroscience Research. ⁺equal contribution. **2020**

Deterioration of grid cells and loss of spatial orientation in Alzheimer's Disease (Cabrera, V.⁺; **Comay, N.A.**⁺; Maza, P.⁺; Vietéz Salvá, E.⁺; Macchione, A.F.⁺). V International Psychology Congress and VII National Psychology Congress "Science and Profession". ⁺equal contribution. **2020**

Teaching experience

GRADUATE-LEVEL

Course instructor and coordinator, *Mathematical and computational modelling of behaviour*, Master's Program in Research Methodology in Behavioural Sciences. **2026**
National University of Córdoba

Teaching assistant, *A practical introduction to cognitive science.* **2023**
National University of Córdoba

UNDERGRADUATE-LEVEL

Teaching assistant, *Tools for data analysis in cognitive neuroscience* **2023, 2026**
National University of Córdoba

Undergraduate teaching assistant, *Neurophysiology and Psychophysiology* **2020-2022**
National University of Córdoba

Undergraduate teaching assistant, *Experimental Psychobiology* **2019-2021**
National University of Córdoba

Human resources training

UNDERGRADUATE STUDENTS SUPERVISED IN THE PSYCHOLOGY PROGRAM

Elias Franco Parras (UNC, Argentina). **2025-2026**

Juan Taravella & Lucia Molina (UNC, Argentina).	2024-2025
Gaspar Taricco (UNC, Argentina). <i>Co-supervised</i> .	2024-2025
Bautista Squire (UNC, Argentina). <i>Co-supervised</i> .	2024-2025
Agostina Presuttari & Camila Abud (UNC, Argentina).	2024-2025
Vera Candusso & Julieta Irazuzta (UNC, Argentina).	2024-2025
Benjamin Sanchez (UNC, Argentina). <i>Co-supervised</i> .	2023-2024
Camila Inés Amado (UNC, Argentina).	2023-2024
Candelaria Nicolas & Florencia Ledesma (UNC, Argentina). <i>Co-supervised</i> .	2023-2024

Peer review and academic service

PEER REVIEW FOR JOURNALS

Nature communications; Nature Machine Intelligence; NeuroImage: Reports; Scientific Reports; npj Artificial Intelligence; Frontiers in Psychology

UNDERGRADUATE THESES EVALUATED

Evaluation of five undergraduate theses in Psychology at the National University of Córdoba.

INSTITUTIONAL SERVICE

Member of the ethics committee for the Use and Care of Laboratory Animals at the Psychological Research Institute (UNC-CONICET) **2022-present**

Skills

Programming (R, MATLAB).

Computational modelling.

Advanced statistical analysis.

Data analysis and visualisation.

fMRI: FSL (FMRIB Software Library) & fMRIPrep.

EEG: EEGLAB.

Selected summer schools & courses

Neuroscience of Consciousness Winter School (Canadian Institute for Advanced Research)	2024
RIKEN Center for Brain Science Summer School (RIKEN CBS, Japan)	2023
Barcelona Summer School for Advanced Modeling of Behavior (Centre de Recerca Matemàtica, Spain)	2022
Statistics applied to health sciences (UNC, Argentina)	2022
First Annual Summer School on Consciousness and Metacognition (University College London & Paris Sciences et Lettres)	2021
Computational Neuroscience (UNSAM, Argentina)	2021
Python programming (UNSAM, Argentina)	2021
Decision making (UBA, Argentina)	2020
Behavioral economics (UBA, Argentina)	2020

References

Pablo Barttfeld, PhD

Independent Researcher at CONICET

Psychological Research Institute (UNC-CONICET), Argentina

cabalangofeld@gmail.com

Guillermo Solovey, PhD

Independent Researcher at CONICET

Universidad Torcuato Di Tella (UTDT), Argentina

gsolovey@gmail.com